Hao Wu

+86 17751232296 | wuhao26@stu.pku.edu.cn | JayLon1022.github.io | github.com/JayLon1022

Education

Peking University Sept. 2026 – Present
M.S., School of Software & Microelectronics *Beijing, China*

Sichuan University Sept. 2022 – Jun. 2026
B.E., School of Cyber Science and Engineering *Chengdu, China*

- Rank: 1 / 161; GPA: 3.85 / 4.0

Publications & Open-Source Software

†Equal contribution.

Juntang Wang†, Yihan Wang†, Hao Wu†, Jiayu Gao†, Shixin Xu, and Dongmian Zou. "StageGuard: Physiologically Constrained Sleep Staging." *KDD 2026*.

Juntang Wang†, Hao Wu†, Yihan Wang†, Dongmian Zou, and Shixin Xu. "Mixing Configurations for Downstream Prediction." *ICML 2026*.

Hao Wu, Jiangchuan Chen, Wengang Ma, Ping He, Xiaolong Lan, Tao Li, and Junjiang He. "LLM-Based Immune Detection Method for Unknown Network Attacks in ICS Under Few-Shot Conditions." *ICIC 2025*.

Juntang Wang†, Yihan Wang†, Hao Wu†, Dongmian Zou, and Shixin Xu. "Brain-Inspired Perspective on Configurations: Unsupervised Similarity and Early Cognition." *BICS 2025*.

mheatmap | Proportional heatmap visualization and matrix analysis with label alignment | 682 Stars | [GitHub]

llmDataG | High-quality tabular data augmentation via LLMs | 40 Stars | [GitHub]

GCN4ETD | Encrypted traffic detection via graph convolutional networks | 11 Stars | [GitHub]

Research & Work Experience

Representation Learning; Physics/Manifold-Informed Neural Networks; Biomedical AI Dec. 2024 – Present
Duke Kunshan University, Division of Natural and Applied Sciences; Research Assistant *Kunshan, China*
Advisors: Prof. Shixin Xu and Prof. Dongmian Zou

- Developed a plug-and-play feature augmentation module for downstream prediction that extracts stable clustering configurations at multiple resolutions and learns sample-specific mixing weights, allowing predictors to use structure at different scales.
- Designed a backbone-agnostic framework for physiologically constrained sleep staging, combining a soft transition penalty during training with duration-aware semi-Markov decoding to discourage implausible transitions and excessively short sleep-stage bouts.
- Contributed to two industry projects: compensating for robotic-arm positioning errors caused by thermal drift, and predicting adhesive-dispensing quality with process-parameter recommendations.

Computer Immunity; Network Intrusion Detection; Noisy-Label Learning Jul. 2023 – Jun. 2026
Sichuan University, Cyber Range Innovation Research Center; Research Assistant *Chengdu, China*
Advisors: Prof. Tao Li and Prof. Junjiang He

- Designed an antigen-adaptive evolution algorithm based on immune gene mechanisms, improving detector diversity and attack-feature coverage.
- For few-shot unknown-attack detection, designed an LLM-optimized artificial immune algorithm: prompt templates guide the model to analyze antigen-space distributions and iteratively generate mutant antibodies, with immune tolerance used to filter detectors.
- Tackled noisy labels in encrypted traffic via a multi-resolution community-structure method, using neighborhood consistency, community support, and cross-granularity stability to identify mislabeled samples, then encoded community configurations as structural features to improve downstream robustness.

RAG-based Question-Answering Service Jun. 2025 – Jul. 2025
Topsec Networks (Beijing), Product R&D Intern *Chengdu, China*

- Built a LangChain + Faiss RAG pipeline covering document chunking, vector indexing, similarity retrieval, and context-augmented generation.

Honors & Competitions

- **Sichuan Province Outstanding Graduate**, 2026.
- **Sichuan Province Class-A Comprehensive Quality Certification**.
- **Ningxia Yanbao Scholarship**, 2025.
- **Sichuan University Outstanding Student Award**, 2023, 2024, 2025.
- **Sichuan University Second-Class Scholarship**, 2023, 2024, 2025.
- **National Project Leader**, National College Student Innovation & Entrepreneurship Program, 2024.
- **National Second Prize**, China Undergraduate Mathematical Contest in Modeling, 2023.
- **Sichuan Second Prize**, China Undergraduate Mathematical Contest in Modeling, 2024.
- **Sichuan Bronze Award**, 9th "Internet+" College Student Innovation & Entrepreneurship Competition.
- **Third Prize**, China Undergraduate Mathematics Competition, 2023.

Additional Information

Service: Grade Leader of the Class of 2022, School of Cyber Science and Engineering, Sichuan University; President of St.808 Student Association; Lead in SCU's "Youthful SCU" promotional music video.
Skills: Python, SQL, C; scikit-learn, PyTorch, NumPy, pandas; Git, LaTeX.
Certification: Sangfor security certification (SCSA-S).